

Python: Fixed Size Warehouse

Develop a `Product` class and a custom exception class to manage a warehouse that can store a limited number of products. If an attempt is made to add a product beyond this limit, it should raise a custom error.

Create the following two classes:

Product

- `__init__(self, name)`: Initializes the product with a name.
- `__del__(self)`: Deletes the product instance.
- Tracks the number of Product instances.
- The maximum number of products can be dynamically changed using `Limit.get_limit()`.
- If the current number of Product instances is below the limit, a new product instance is created.
- If adding a new product exceeds the limit, raise a `UserLimitExceeded` exception with the message "Product {name} cannot be created. Maximum {limit} products allowed" and do not create the new instance.

UserLimitExceeded(Exception)

- This is a custom exception class.
- It accepts an error message in its constructor and raises an exception.

****Note:**** The template provides the `Limit` class and its operations. Review this code to understand how it integrates with the application.

Function Description

Complete the classes `Product` and `UserLimitExceeded` in the editor.

Instructions are handled by the boilerplate code.

Constraints

- The maximum number of instructions is 100.
- The maximum number of products is 50.
- The limit will never be changed to less than the current number of Product instances.

Input Format For Custom Testing

The first line contains an integer, `n`, the initial limit of `Product` instances.

The next line contains an integer `m`, the number of operations.

Each of the next `m` lines contains an instruction in the format `command argument`, where `command` is one of `[new, del, limit, print]` and `argument` is the value.

Sample Case 0

Sample Input For Custom Testing

STDIN	Function
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```

-----
3          limit n = 3
10         num ops m = 10
new laptop ops = ["new laptop", "new bag", "new LCD", "print laptop", "new keyboard",
"del laptop", "new chair", "del keyboard", "print keyboard", "del laptop"]
new bag
new LCD
print laptop
new keyboard
del laptop
new chair
del keyboard
print keyboard
del laptop

```

Sample Output

```

laptop created
bag created
LCD created
laptop found
Product keyboard cannot be created. Maximum 3 products allowed
laptop deleted successfully
chair created
keyboard not found
laptop not found

```

Explanation

Initially, the limit is 3. New products are created. When the limit is reached, the new product is not added and an error is thrown. Note that once a product is deleted, it is no longer found.

Sample Case 1

Sample Input For Custom Testing

```

STDIN      Function
-----
1          limit n = 1
6          num ops m = 6
new desktop ops = ["new desktop", "new car", "limit 2", "new car", "print car", "print
truck"]
new car
limit 2
new car
print car
print truck

```

Sample Output

```

desktop created
Product car cannot be created. Maximum 1 products allowed

```

```
limit updated to 2  
car created  
car found  
truck not found
```

Explanation

Initially, the limit is 1. When the limit is updated, new products can be added.